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CSD 380 Module 1.3 Assignment: The History of DevOps
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Repository: https://github.com/fernandojosecc/csd380-devops

The History of DevOps

Introduction

DevOps did not appear all at once. It grew out of three earlier movements, Lean, Agile, and Continuous Delivery, that each tried to deliver value faster with fewer problems. The DevOps Handbook describes DevOps as a convergence of many management and technology movements. This paper follows that history through the three movements and ends with the 2009 events where Development and Operations ideas came together and the term “DevOps” was coined.

The Lean Movement

Lean is the oldest of the three and did not start in software. It grew out of the Toyota Production System (TPS), which Toyota says rests on just-in-time, an idea from founder Kiichiro Toyoda, and jidoka, which came from Sakichi Toyoda’s work on automatic looms. Taiichi Ohno applied both ideas to making cars. Toyota built the system through trial and error starting in the late 1940s and began rolling it out to all plants in 1960. The Lean Enterprise Institute notes that its creators had no grand plan, and the system grew through decades of experimentation.

By the 1980s, value stream mapping, kanban boards, and total productive maintenance had been codified for TPS, and in 1997 the Lean Enterprise Institute began applying Lean to services and healthcare. The institute describes lean as “a way of thinking about creating needed value with fewer resources and less waste”. Two Lean beliefs matter most for DevOps: lead time is the best predictor of quality, customer satisfaction, and employee happiness, and small batch sizes are one of the best predictors of short lead times.

The Agile Manifesto

The Agile Manifesto answered a different problem: software projects that moved slowly and could not handle change. From February 11 to 13, 2001, seventeen people from methods such as Extreme Programming, Scrum, and Crystal met at the Snowbird ski resort in Utah. They worked in what was then called “lightweight methods,” but they disliked the word “light” and chose “Agile” instead. Their goal was to capture the advantages of adaptive methods over waterfall development and heavier processes like the Rational Unified Process.

The Manifesto states four values: individuals and interactions over processes and tools, working software over comprehensive documentation, customer collaboration over contract negotiation, and responding to change over following a plan. The authors add that while there is value in the items on the right, they value the items on the left more. One principle was to “deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the

shorter timescale". That is a call for small batches. It matters for DevOps because many key DevOps moments happened in the Agile community, such as a 2008 session in Toronto on applying Agile ideas to infrastructure.

The Continuous Delivery Movement

Continuous Delivery built on the practice of continuously building, testing, and integrating code. Jez Humble and David Farley extended it with the deployment pipeline, which keeps code and infrastructure always in a deployable state so that anything checked into trunk can be safely deployed to production. The idea was first presented at the 2006 Agile conference. Their book, *Continuous Delivery*, came out in July 2010. It defines the pipeline as "an automated process for managing all changes, from check-in to release," with the goal of releasing small changes frequently, even within hours or minutes.

Martin Fowler explains the idea simply: the current version of the software can be deployed to production at a moment's notice. He also separates two terms. In continuous deployment, every change that passes the pipeline goes to production automatically. In continuous delivery, the team can deploy often but may choose not to. This movement mattered to DevOps because it made release and deployment part of one automated flow shared by Development and Operations.

How the Movements Became DevOps

By the late 2000s the pieces were in place, but Development and Operations were still rarely working on them together. In 2008, Patrick Debois and Andrew Shafer held a session at the Agile conference in Toronto on applying Agile principles to infrastructure. They were the only two who showed up, but they soon gained a following that included John Willis. In 2009, at the Velocity conference, John Allspaw and Paul Hammond presented "10 Deploys per Day: Dev and Ops Cooperation at Flickr," showing how shared goals and continuous integration made deployment part of everyone's daily work. Debois was so excited by the talk that he created the first DevOpsDays in Ghent, Belgium, in 2009, where the term "DevOps" was coined. The book sums up the result as applying Lean principles to the technology value stream through the Three Ways: Flow, Feedback, and Continual Learning and Experimentation.

Conclusion

DevOps is the meeting point of three movements. Lean taught that short lead times and small batches predict quality. Agile showed that short cycles and close collaboration beat long plans. Continuous Delivery showed how automation can keep software ready to release at any time. Together they point at one goal: getting valuable work to customers quickly and safely.

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